Response to Referee

Necessary and Sufficient Conditions for the Existence of an LU Factorization for General Rank Deficient Matrices

Dear Editor,

We are grateful to the referee for the careful reading of the manuscript and for the detailed comments. The revised manuscript addresses the issues raised in the report. In particular, we have clarified the induction steps in the main proofs, expanded the discussion of the relationship with existing results, and revised several points of notation and exposition.

Below we give a point-by-point response. Referee comments are shown in italics. Our responses follow each comment.

Summary of Main Changes

- [Will do at the end]

Main Points

Main Point 1

Referee comment.

> *The new results (in fact, examples) only start from Page 5, and the theoretical contribution does not appear to justify the current length. In my opinion, the manuscript can be substantially cut down by providing only references to known results and, with a proper literature review, emphasizing on how the new theorem improves on existing results.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 2

Referee comment.

> *The referee asks that the proof of Theorem 1.1 spell out how the permutation and the factorization of the Schur complement are embedded into the full matrix. The referee also asks for the analogous induction step in Theorem 1.4 and for explicit mention of the Schur complement.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 3

Referee comment.

> *The referee asks for an explanation of why the construction terminates after $r = \operatorname{rank}(A)$ steps, and suggests that the later statement in Theorem 2.5 can refer back to this explanation.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 4

Referee comment.

> *The referee notes that Property 2.1 is cited as Theorem 2.1, and questions the separation of statements and proofs for the main results.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 5

Referee comment.

> *The referee asks for more detail in the proof of Theorem 2.5 explaining why the intermediate matrix D and the Schur complement S inherit the same rank condition.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 6

Referee comment.

> *The referee asks whether the linear independence argument still holds in Case 1 of the proof of Theorem 2.5, since it is no longer clear that $B(1, 2 : k) = 0$ for $k \geq j$.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Main Point 7

Referee comment.

> *The referee points out that the necessary direction of Theorem 3.1 can be proved more directly by using*
>
> $$A(:, 1 : k) = L(:, 1 : k)U(1 : k, 1 : k), \qquad A(1 : k, 1 : k) = L(1 : k, 1 : k)U(1 : k, 1 : k),$$
>
> *and the full column rank of $L(:, 1 : k)$ and $L(1 : k, 1 : k)$.*

Response. [Write the response here.]

Change made. [Describe the manuscript change and give page, theorem, equation, or line references.]

Minor Points

Minor Point 1: Singular and rank-deficient

Referee comment.

> *For square matrices, the terms "singular" and "rank-deficient" are interchangeable. The referee suggests keeping only one where applicable.*

Response. [Write the response here.]

Minor Point 2: Punctuation of displayed mathematics

Referee comment.

The referee asks that mathematical expressions be treated as part of the sentence and punctuated accordingly.

Response. [Write the response here.]

Minor Point 3: More formal row and column phrasing

Referee comment.

The referee suggests replacing expressions such as "column i" and "row j" by "the ith column" and "the jth row".

Response. [Write the response here.]

Minor Point 4: Matrix notation

Referee comment.

The referee suggests replacing MATLAB-style notation such as $A[:, i]$ by standard numerical linear algebra notation, and using block notation for submatrices.

Response. [Write the response here.]

Minor Point 5: References for applications

Referee comment.

The referee asks for references in the discussion of the physical setting and application background on page 2.

Response. [Write the response here.]

Minor Point 6: Definition of A_{11}

Referee comment.

The referee notes that A_{11} appears on page 2 without a proper definition.

Response. [Write the response here.]

Minor Point 7: Wording in the proof of Theorem 1.1

Referee comment.

> *The referee suggests changing the second line of Case 1 in the proof of Theorem 1.1 to say that B has a partially pivoted LU factorization.*

Response. [Write the response here.]

Minor Point 8: Proof of Corollary 1.2

Referee comment.

> *The referee suggests writing: "Apply Theorem 1.1 to A^T and take the transpose of the result."*

Response. [Write the response here.]

Minor Point 9: Definition 1.3

Referee comment.

> *The referee suggests replacing "triangular matrix" by "trapezoidal matrix" in Definition 1.3.*

Response. [Write the response here.]

Minor Point 10: Colon after "Therefore"

Referee comment.

> *The referee notes that in line 3 of the proof of Proposition 1.5, the colon after "Therefore" should be a comma.*

Response. [Write the response here.]

Minor Point 11: Theorem 2.2–Corollary 2.4

Referee comment.

> *The referee suggests replacing this material by references to the literature, for example Exercise 4.5.10 in Carl D. Meyer's* Matrix Analysis and Applied Linear Algebra.

Response. [Write the response here.]

Minor Point 12: Key property from the smallest index j

Referee comment.

> *The referee notes that in the proof of Theorem 2.5, the key property following from the smallest index j is that $B(1, 2 : k) = 0$, and asks that this be mentioned for readability.*

Response. [Write the response here.]

Minor Point 13: "non zero"

Referee comment.

> *The referee notes that "non zero" should be written as "nonzero."*

Response. [Write the response here.]

Minor Point 14: "B satisfies Case 2"

Referee comment.

> *The referee notes that the phrase "B satisfies Case 2" is imprecise because Case 2 is stated for A.*

Response. [Write the response here.]

Minor Point 15: Rank-revealing factors L_B and U_B

Referee comment.

> *The referee found it unclear whether the standalone paragraphs about the rank-revealing factors L_B and U_B are part of Theorem 2.6.*

Response. [Write the response here.]

Minor Point 16: Typos in sparsity expressions

Referee comment.

> *The referee notes typos in expressions such as $L[i_0, i + 1 :]$ and $L[i_0, i_0 :]$, and similar errors elsewhere.*

Response. [Write the response here.]

Minor Point 17: Figure for Theorem 2.6

Referee comment.

> *The referee suggests adding a graphical illustration of the sparsity pattern of the LU factors.*

Response. [Write the response here.]

Minor Point 18: "an index j_0"

Referee comment.

> *The referee suggests replacing "We also have a j_0 such that..." by "an index j_0 such that..."*

Response. [Write the response here.]

Minor Point 19: Row and column permutations in the unit lower case

Referee comment.

> *The referee comments on the sentence "The triangular unit structure of L prevents row permutations" and suggests clarifying that no row permutation can be performed to A.*

Response. [Write the response here.]

Minor Point 20: Table 3

Referee comment.

> *The referee notes that the column and row conditions in Table 3 appear to be with respect to A_{11}, and asks that this be made clear.*

Response. [Write the response here.]

Minor Point 21: Paragraph above Lemma A.3

Referee comment.

The referee suggests explicitly stating

$$(I - P_o(X))A = \begin{pmatrix} O_{k-1} & O \\ O & S \end{pmatrix}$$

and adding some discussion.

Response. [Write the response here.]

Minor Point 22: Lemmas A.1–A.3

Referee comment.

> *The referee suggests moving Lemmas A.1–A.3 into the main text if the results are new.*

Response. [Write the response here.]

Closing

We hope that the revisions address the referee's concerns and make the manuscript clearer and more useful to readers.